Large Language Models for Social Science Research

Session 1. Setting Up an Agentic Research Project

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PhD Workshop, University of Zurich

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Before We Start

9:00 – 9:20

The idea of this session, and why it is debatable

An Economics Paper, Produced Autonomously

- An AI system writes economics papers on its own: question, data, code, results, text.
- Autonomous Policy Evaluation (APE), Social Catalyst Lab of David Yanagizawa-Drott, University of Zurich.
- 900+ papers by spring 2026, ranked by LLM referees against published papers.

▶ Video, 2 minutes

Posted on LinkedIn by the project's author, February 2026

The Author's Own Reading: Two Worlds

World 1. This is real research. But it has nothing to do with my expertise. I cannot evaluate it. Some other human expert has to.

World 2. This is not real research. I just produced a negative externality by polluting the information environment.

His conclusion. In a world of machine-speed papers, we have a problem either way. Generation speed $\gg$ verification speed. This was never the case historically. Anybody, with no expertise, can now produce a paper that looks legitimate.

And two questions he leaves open: if anyone can produce a paper, how do we tell who is good, and who gets the credit? "I don't have the answers."

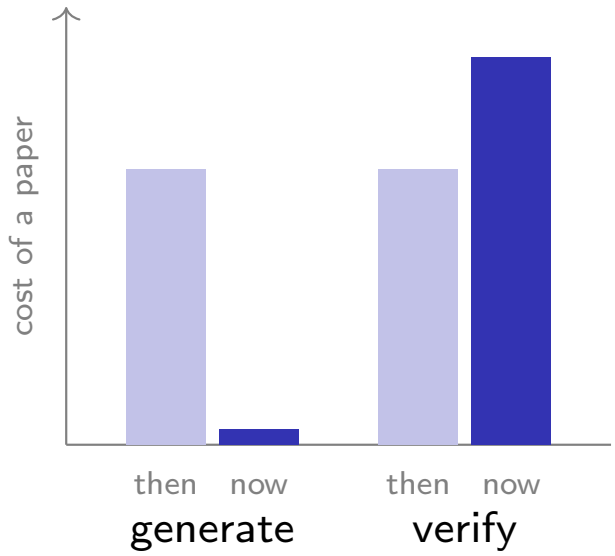
Neither do I. But we can push this a little further...

Pushing Further (1): The Bottleneck Moved

His two worlds have one consequence in common. Either way, the scarce resource is no longer production. It is qualified verification. The bottleneck moved from *doing* to *judging*.

Why did this never happen before?

- Verification was always slow. But generation was slow too.
- The cost of writing a paper was a filter: you did not write one you did not believe in.
- It was also a signal of competence (Spence 1973): costly, hence informative.
- Peer review was calibrated for a world where submitting cost a year.



Generating: almost free.

Verifying: harder than before.

Pushing Further (2): Verification at Machine Speed

The APE papers have already been checked from the outside.

- **Cunningham (2026)**, 651 papers: systematic p-hacking.
- **Ning (2026)**, 912 papers against 41 from the *AER* and *AEJ*: the human papers win 83% of comparisons. The gap is in the *idea*, not in the execution.

Both checks were run with LLMs. So is the APE tournament.

Who verifies the verifier?

The chain has to end with a human who understands.

Pushing Further (3): What You Cannot Delegate

The skill to check. It is built by doing. Stop doing, and within one cohort nobody can check any more. Everyone then depends on whoever still can.

During Covid, countries that had stopped making masks and paracetamol could not make them when it mattered.

The accountability. You can delegate a task, not the responsibility for it. If the agent is wrong, it loses nothing: no reputation, no retraction, no referee to answer. Your name is on the paper.

The senior PI who lets the RAs do everything and stops checking: when it breaks, it is still the PI's paper.

Delegate the work. Keep the skill to check it,
and the responsibility for it.

The Trade-off, and Why It Is Yours to Make

Short term, individual: publish or perish. Delegating everything is rational: committees count papers.

Long term, collective. Expertise is a common-pool resource: every person who stops understanding depletes it a little.

But is publish or perish from another age?

- When anyone can produce a paper, a paper stops being a signal.
- What stays scarce: the idea, and the ability to vouch for the result.
- Not tomorrow: committees still count papers.
- You are hit first: agents take over the tasks you learn on.

Feeling productive is not being productive: in a randomised trial run in early 2025 (Cursor with Claude 3.5/3.7 Sonnet), experienced developers were 19% slower with AI, while believing they were 20% faster (Becker et al. 2025).

I am not going to pretend this trade-off is balanced.
It is your choice. Here is mine.

My Position, and Three Rules for This Workshop

Where I stand

I am not here to teach you the technique. I am here to show you how to delegate it well. You stay accountable for what you delegate, so you have to understand it.

1. **You delegate execution, never understanding.** You must be able to explain every file in your project.
2. **Read before you accept.** Every output is a draft. Every session has a mandatory manual check.
3. **Stop when you no longer know what it is doing.** Fatigue is the signal to stop, not to accept faster.

The frontier is jagged: just outside it, consultants with GPT-4 were 19% *less* likely to get a task right than without it (Dell'Acqua et al. 2023).

Knowing where the frontier lies is a skill, and you learn it by doing.

Three Costs We Will Not Study Here

Environmental. Data centres: 415 TWh in 2024, 1.5% of world electricity, more than double by 2030 (IEA 2025). One prompt is small, about nine seconds of TV (Google 2025), but an agent chains many.

Human, collective. Behind the models, low-paid workers. In Kenya, people labelled violent and sexual content for ChatGPT's safety filter for \$1.32 to \$2 an hour (Perrigo 2023).

More in my course on AI: sites.google.com/view/hugosubtil/teaching.

Personal. Supervising is tiring: 77% of employees who use AI say it added to their workload (Upwork 2024). Losing ownership of your work erodes the sense of accomplishment, a core dimension of burnout (Maslach & Leiter 2016). Take care of yourselves.

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This is why we start with an idea, not with a chat.
And we put it in a folder: a project you own, read and understand.

Today's Session Overview (9:00–12:00)

9:00–9:20 Before We Start (20 min)

9:20–9:35 An Agentic Research Project (15 min)

- Chat vs agent, why a folder, the template, guardrails

9:35–9:50 Installation Clinic (15 min)

- Worksheet, steps 0 and 1

9:50–10:25 Live Demo (35 min), then a 15-min break

- From an empty folder to a first commit

10:40–11:40 Hands-on (60 min)

- Your project: worksheet, steps 1 to 8

11:40–12:00 Debrief and Next Session (20 min)

An Agentic Research Project

9:20 – 9:35

From a chat that forgets to a folder that remembers

Course Repository



github.com/subtilhugo/uzh-llm-socsci-2026

Chat vs Agent

A chat

- You paste, it answers, you copy. It sees nothing else.
- It forgets everything when you close the tab.
- It cannot open your PDF, run your script, or look at your data.

An agent

- It works inside a folder on your machine.
- The loop: read files $\rightarrow$ run commands $\rightarrow$ write files $\rightarrow$ report.
- It asks permission before each action that changes something.
- Claude Code, Gemini CLI, Codex: same idea.

Same model as in the chat. What changes is the loop, the tools, and the permission prompt.

Why a Folder, Not a Chat

1. **Context.** The agent reads `CLAUDE.md` at every start. It grows with the project: you edit it, or ask the agent to. A long chat degrades; a fresh start does not.

2. **Memory.** `PROJECT_HISTORY.md` says where the project stands. A fresh session can answer "where are we?" from the file alone. Nothing lives in the chat.

3. **Reproducibility.** Git, versioned prompts, scripts that regenerate outputs from raw data. In a year you will know what was done, from what, and by whom.

And one more: the folder is yours. Change tool tomorrow, the folder stays: the same `CLAUDE.md` works in the terminal, Cursor or the desktop app. A chat is the vendor's; the folder is yours.

Anatomy of the Project Template

`my-project/`	
`README.md`	what the project is
`CLAUDE.md`	how the agent behaves
`PROJECT_HISTORY.md`	where we stand
`literature/`	PDFs (read-only)
`notes/literature/`	one note per paper
`notes/`	memos, critiques
`data/raw/`	never modified
`data/processed/`	rebuilt by scripts
`code/`	numbered scripts
`outputs/`	figures, tables
`prompts/`	the prompts, versioned
`paper/`	drafts
`.gitignore`	`.env`, data, PDFs

Three files carry everything.

- `README.md`: question, design, data, people.
- `CLAUDE.md`: rules for the agent, your conventions.
- `PROJECT_HISTORY.md`: done, decided, *verified*, open.

Guardrails

Written in `CLAUDE.md`

- `data/raw/` is read-only. Always.
- Never invent a reference: not in the folder = "not found".
- Never delete, never push. Every claim cites file and page.

Written in your habits

- What the agent reads is sent to the vendor's servers. What it does not open stays on your machine.
- Confidential data: show the agent a sample or synthetic data; its script then runs on the real file, unopened.
- API keys live in `.env`; `.env` is in `.gitignore`.
- Stay in "ask" mode: it proposes, you approve.

A rule in a file is a request, not a guarantee.

What You Leave With at Noon

A git folder for your project, containing:

- `README.md` with your question, in your words
- `CLAUDE.md` with your conventions
- 3 to 5 literature notes, one verified by you against the PDF
- a one-page research memo
- a hostile critique of that memo, and your response
- an entry in `PROJECT_HISTORY.md` and a first commit

And one thing you will have seen: the agent making a mistake in a literature note. A wrong number, a wrong page, a finding that is not in the paper. Finding it is step 5 of the worksheet. It is the only step you cannot skip.

No project yet, or no PDFs today? Use the example project. You lose nothing.

Installation Clinic

9:35 – 9:50

Worksheet, step 0, then step 1

Stuck? Paste the error in the shared document and ask your neighbour.

Live Demo

9:50 – 10:25

Who votes far right? The local rise of the FN/RN

The Demo in Seven Steps

1. Empty folder, template, `git init`, start the agent
2. Fill the context (prompt 01)
3. Literature notes from the PDFs (prompt 02)
4. Verify a note against the PDF
5. Research memo (prompt 03)
6. Hostile reviewer (prompt 04)
7. Log, commit, close; ask a fresh session where we stand (prompt 05)

All prompts are in `template/prompts/`. You will paste the same ones on your project.

Hands-on

10:40 – 11:40

Worksheet, steps 1 to 8. Your project, your PDFs.

Steps 1–3 first. Step 5 is mandatory.

Stuck for five minutes? Paste the error in the shared document and ask your neighbour.

Debrief

Three screens. A `CLAUDE.md`. A literature note, with the error that was found. A critique, with the response.

The errors we found today. Read from the shared document. The list is the lesson.

One question. Would you have caught that error in a paper you did not know? That is the gap between generation and verification, on your own desk.

Before you leave. One line in your `CLAUDE.md` that would have prevented it.

Next Session: 28 October 2026

Session 2: Structured Data with an Agent

- Bring your dataset as a file the agent can open (CSV, Excel, Stata, SPSS, Parquet). Confidential: a sample. Large: a subset.
- No dataset: you will use the example pack (French election results and local indicators, by municipality).
- Keep working in your folder. End every session with prompt 05.

Carry-over question

Next time the agent will clean and merge your data. How would you know that a merge it wrote silently dropped a tenth of your observations?

References

Becker, J., Rush, N., Barnes, E., & Rein, D. (2025). Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity. *METR*, arXiv:2507.09089.

Cunningham, S. (2026). Do AI Agents Writing Full Manuscripts at the Social Catalyst Lab P-Hack? *causalinf.substack.com*.

Dell'Acqua, F., McFowland III, E., Mollick, E., et al. (2023). Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality. *Harvard Business School Working Paper* 24-013.

Google (2025). Measuring the Environmental Impact of AI Inference. *Google Cloud Blog*, August 2025.

IEA (2025). *Energy and AI*. International Energy Agency, Paris.

Maslach, C. & Leiter, M. P. (2016). Understanding the Burnout Experience: Recent Research and Its Implications for Psychiatry. *World Psychiatry*, 15(2), 103–111.

Ning, L. (2026). The Ideation Bottleneck: Decomposing the Quality Gap Between AI-Generated and Human Economics Research. arXiv:2604.03338.

Perrigo, B. (2023). OpenAI Used Kenyan Workers on Less Than $2 Per Hour to Make ChatGPT Less Toxic. *TIME*, 18 January.

Spence, M. (1973). Job Market Signaling. *Quarterly Journal of Economics*, 87(3), 355–374.

Upwork Research Institute (2024). Survey of 2,500 workers in the US, UK, Australia and Canada. Press release, 23 July 2024.